

Request for Comment

Dune Discord Bot & Adapter Multi-Train Release Roadmap

Status: Draft

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Repository: yacketrj/dune-awakening-selfhost-discordbot

Core Adapter: Red-Blink/dune-awakening-selfhost-docker

1. Abstract

This RFC defines a multi-train release roadmap for the Dune Awakening Selfhost Discord Bot and its companion WebUI adapter. The roadmap delivers read-only operational visibility first, write-safety foundation second, and controlled game-facing operations only after lower-risk trains are proven in production.

The bot operates as an external companion to the Dune Docker Console WebUI. All commands go through a bearer-token protected adapter API. At no point does the bot access the Docker socket, Postgres database, game files, or shell commands directly.

2. Motivation

Operators and server administrators need a safe way to monitor and manage their Dune Awakening dedicated servers through Discord. The bot must:

- Provide read-only operational visibility (server status, readiness, services, population).
- Support scheduled status posts, alert subscriptions, and incident digests.
- Eventually support administrative writes (maintenance notes, notification config, scheduled posts).
- Eventually support operational writes (backup creation, service restart, update trigger).
- Support game-to-Discord announcement bridging and Discord-to-game broadcast commands under careful authorization gates.
- Never expose raw database access, shell execution, Docker socket access, or game-server command passthrough through Discord.

3. Design Principles

1. **Security gates first; functionality second.** Every PR passes npm audit, Semgrep SAST, Gitleaks, Trivy filesystem and image scans, dependency review, and test gates before merge.

2. **Read-only by default.** All commands are read-only until the R2.0.0 write-safety foundation is approved and upstream write-contract is in place.
3. **Write paths disabled by default.** Every write command requires `DUNE_DISCORD_WRITES_ENABLED=true` and write-specific RBAC that observer roles do not inherit.
4. **Confirmation cannot be bypassed.** Write commands require explicit confirmation with action, target, and risk displayed.
5. **All output is bounded, redacted, and role-gated.** PII, secrets, internal IPs, Steam IDs, Funcom IDs, and credentials are redacted before Discord output.
6. **No direct infrastructure access.** The bot never mounts the Docker socket, writes to Postgres, stores secrets in addon files, or executes shell commands.
7. **Zero-permission addon.** The WebUI addon is a static pointer with no backend access.

4. Release Train Model

Train	Scope	Key Commands	Status
R1.0.0 (v1.0.0)	Stable read-only GA	about, ping, health, status, status-summary, readiness, services	Implemented
R1.1-R1.5 (v1.1.0-v1.5.0)	Read-only maturity	population, backups, scheduler, validation, compatibility	Implemented
R2.0.0 (v2.0.0)	Write-safety foundation	Disabled-by-default write framework, RBAC, idempotency, audit	Feature Branch
R2.1-R2.4 (v2.1.0-v2.4.0)	Low-risk admin writes	Maintenance notes, notification config, scheduled-post config	Feature Branches
R3.0.0 (v3.0.0)	Operational writes	Backup creation, service restart, update trigger, cache clear	Feature Branch
R4.0.0+ (v4.0.0+)	Highest-risk operations	Player moderation, game broadcast, chat bridge, restore execution	Feature Branches

5. Security Gates

Every substantive PR, release candidate, and stable release must pass the following gates:

- `npm run check` — 125 unit tests + release metadata + addon package + SBOM
- `npm audit --audit-level=moderate` — zero vulnerabilities
- Semgrep SAST — zero findings
- Gitleaks secret scan — zero leaks (with documented false-positive allowlist)
- Trivy filesystem scan — zero findings
- Trivy image scan — zero findings
- Docker build — success
- Dependabot cooldown (7-day default) — prevents supply-chain poisoning
- Pinned GitHub Actions (immutable SHA) — prevents workflow injection
- No unresolved medium, high, or critical security findings

6. Communication Bridge

6.1 Game → Discord (Announcements)

In-game announcements (broadcasts, server events, shutdown warnings) are forwarded to a configured Discord channel. The bridge polls the adapter for new announcements at configurable intervals, deduplicates by ID, and caps message length at 1800 characters. All output is redacted before posting.

Disabled by default: `DUNE_ANNOUNCEMENTS_ENABLED=true` to enable.

6.2 Discord → Game (Broadcast Command)

Moderator+ roles can use `/dune broadcast <message>` to send a message to all in-game players. The command requires write-enabled mode, RBAC check, confirmation, idempotency key, 60-second per-user cooldown, and produces a tamper-evident audit event.

Disabled by default: `DUNE_DISCORD_WRITES_ENABLED=true` and write-role assignment required.

7. Current Implementation Evidence

Metric	Value
Tests passing	125 / 125
npm audit	0 vulnerabilities
Semgrep	0 findings
Gitleaks	0 leaks
Trivy filesystem	0 findings
Trivy image	0 findings
Commands	10 (about, ping, health, status, status-summary, readiness, services, population, backups, broadcast)
Source modules	12 (adapterClient, announcements, broadcast, commands, config, cooldown, format, healthState, index, logger, notifications, scheduler)
RBAC mode	restricted by default, fails closed
Addon	zero-permission, static HTML
Container	non-root (USER node), HEALTHCHECK, npm removed from runtime

8. Appendix — Source References

- [docs/full-release-roadmap.md](#) — Detailed release train specification
- [docs/r1-r2-release-roadmap.md](#) — R1.x to R2.x cadence and entry criteria
- [docs/additional-features-roadmap.md](#) — 16 proposed additional features
- [docs/security-audit/2026-07-04-comprehensive-security-audit.md](#) — Full security audit
- [docs/pr-transparency-template.md](#) — PR documentation requirements
- [releases/index.md](#) — Upstream PR staging manifest

RFC — Dune Discord Bot Roadmap • Draft 2026-07-04 • No upstream PRs created • All work is local feature branches staged in releases/